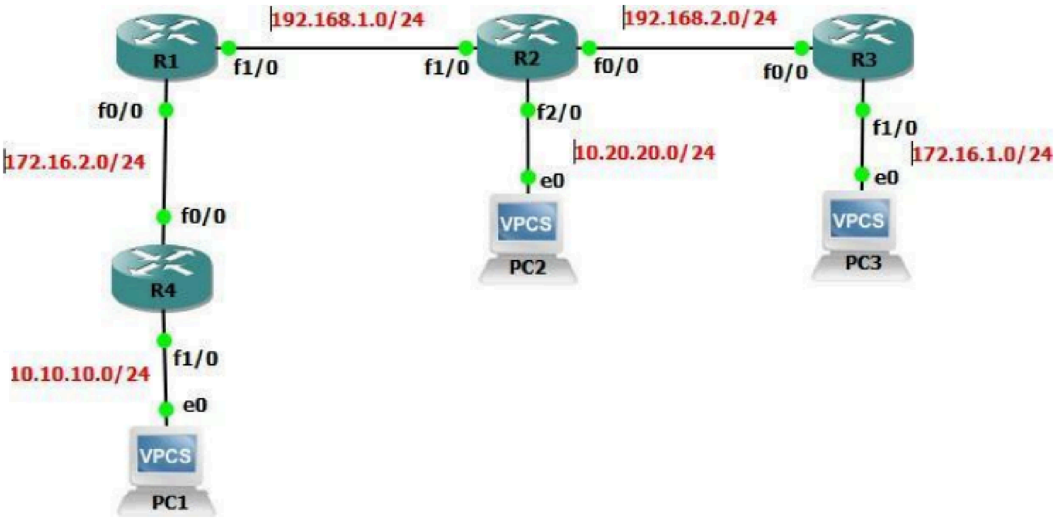
TP1 - R302 BGP

On réalise ce schéma :



Configuration IP :

On commence par mettre les adresses sur tout les équipements.

Configuration IP Routeur 1 :

```
en
!
conf t
!
interface FastEthernet0/0
 ip address 172.16.2.1 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 192.168.1.1 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

Configuration IP Routeur 2 :

```
en
!
conf t
!
interface FastEthernet0/0
 ip address 192.168.2.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 192.168.1.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet2/0
 ip address 10.20.20.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

Configuration IP Routeur 3 :

```
en
!
conf t
!
interface FastEthernet0/0
 ip address 192.168.2.3 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 172.16.1.3 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

Configuration IP Routeur 4 :

```
en
!
conf t
!
interface FastEthernet0/0
```

```
 ip address 172.16.2.4 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 10.10.10.4 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

Configuration IP PC1 :

```
ip 10.10.10.5/24 10.10.10.4
```

Configuration IP PC2 :

```
ip 10.20.20.5/24 10.20.20.2
```

Configuration IP PC3 :

```
ip 172.16.1.5/24 172.16.1.3
```

Configuration eBGP :

On configure le eBGP sur tout les routeurs.

Configuration BGP Routeur 1 :

```
router bgp 65101
!
neighbor 192.168.1.2 remote-as 65102
!
neighbor 172.16.2.4 remote-as 65100
!
network 192.168.1.0 mask 255.255.255.0
!
network 172.16.2.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 2 :

```
router bgp 65102
!
neighbor 192.168.1.1 remote-as 65101
!
neighbor 192.168.2.3 remote-as 65103
!
network 192.168.1.0 mask 255.255.255.0
!
network 192.168.2.0 mask 255.255.255.0
!
network 10.20.20.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 3 :

```
router bgp 65103
!
neighbor 192.168.2.2 remote-as 65102
!
network 192.168.2.0 mask 255.255.255.0
!
network 172.16.1.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 4 :

```
router bgp 65100
!
neighbor 172.16.2.1 remote-as 65101
!
network 172.16.2.0 mask 255.255.255.0
!
network 10.10.10.0 mask 255.255.255.0
!
```

Tout ping bien PC1 peut bien ping PC3 :

```
PC1> ping 172.16.1.5
84 bytes from 172.16.1.5 icmp_seq=1 ttl=60 time=121.611 ms
84 bytes from 172.16.1.5 icmp_seq=2 ttl=60 time=121.224 ms
84 bytes from 172.16.1.5 icmp_seq=3 ttl=60 time=120.784 ms
84 bytes from 172.16.1.5 icmp_seq=4 ttl=60 time=120.742 ms
84 bytes from 172.16.1.5 icmp_seq=5 ttl=60 time=121.637 ms
```

Configuration iBGP :

On configure le iBGP sur R1, R2 et R4.

Configuration iBGP sur R1 :

```
router bgp 65101
!
neighbor 192.168.1.2 remote-as 65101
!
neighbor 172.16.2.4 remote-as 65101
!
network 192.168.1.0 mask 255.255.255.0
!
network 172.16.2.0 mask 255.255.255.0
!
```

Configuration iBGP sur R2 :

```
no router bgp 65102
!
router bgp 65101
!
neighbor 192.168.1.1 remote-as 65101
!
neighbor 192.168.2.3 remote-as 65103
!
neighbor 10.30.30.4 remote-as 65101
!
network 192.168.1.0 mask 255.255.255.0
!
network 192.168.2.0 mask 255.255.255.0
!
network 10.20.20.0 mask 255.255.255.0
!
network 10.30.30.0 mask 255.255.255.0
!
end
!
conf t
!
int f3/0
!
ip address 10.30.30.2 255.255.255.0
!
no sh
!
```

Configuration eBGP sur R3 :

```
router bgp 65103
!
neighbor 192.168.2.2 remote-as 65101
!
network 192.168.2.0 mask 255.255.255.0
!
network 172.16.1.0 mask 255.255.255.0
!
```

Configuration iBGP sur R4 :

```
no router bgp 65100
!
router bgp 65101
!
neighbor 172.16.2.1 remote-as 65101
!
neighbor 10.30.30.2 remote-as 65101
!
network 172.16.2.0 mask 255.255.255.0
!
network 10.10.10.0 mask 255.255.255.0
!
network 10.30.30.0 mask 255.255.255.0
!
end
!
conf t
!
int f2/0
!
ip address 10.30.30.4 255.255.255.0
!
no sh
```

Bloquage réseau PC1 depuis R2 :

On veut empecher R2 de trouver le réseau 10.10.10.0.

Configuration Routeur R4 :

```
route-map BLOCK-PC1 deny 10
!
match ip address 10
!
access-list 10 deny 10.10.10.0 0.0.0.255
!
router bgp 65101
```

```
!
neighbor 10.30.30.2 route-map BLOCK-PC1 out
```

Choix du Chemin : ATTENTION NE MARCHE PAS

On configure R3 pour qu'il choisisse la route

Configuration R3 :

```
ip prefix-list NET_PC1 seq 5 permit 10.10.10.0/24
!
route-map PREF permit 10
!
match ip address prefix-list NET_PC1
!
set local-preference 200
!
exit
!
router bgp 65101
!
neighbor 10.30.30.2 route-map NET_PC1 out
```